

Sum and Difference Formulas

ANSWER KEY



Exact values with sum and difference formulas

- 1 Use a sum or difference formula to find the exact value of $\cos 75^\circ$.
 $\cos 75^\circ = \cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$.
- 2 Find the exact value of $\sin \frac{\pi}{12}$.
 $\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$.
- 3 Find the exact value of $\tan 105^\circ$.
 $\tan 105^\circ = \tan(60^\circ + 45^\circ) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(1 - \sqrt{3})(1 + \sqrt{3})} = \frac{4 + 2\sqrt{3}}{-2} = -2 - \sqrt{3}$.

Working with given values

- 4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:
a $\sin(A + B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.
b $\cos(A + B)$ $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.
c the quadrant of $A + B$ Sine positive, cosine negative: Quadrant II.
- 5 Use a sum or difference formula to simplify:
a $\sin \left(x + \frac{\pi}{2} \right) = \cos x$ **b** $\cos(x - \pi) = -\cos x$
- 6 Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.
In QII, $\cos A = -\frac{15}{17}$ and $\cos B = -\frac{4}{5}$.
 $\cos(A - B) = \cos A \cos B + \sin A \sin B = \left(-\frac{15}{17} \right) \left(-\frac{4}{5} \right) + \frac{8}{17} \cdot \frac{3}{5} = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$.

Verifying identities using sum and difference formulas

- 7 Verify the identity $\cos(x + y) + \cos(x - y) = 2\cos x \cos y$.
 $\cos(x + y) = \cos x \cos y - \sin x \sin y$ and $\cos(x - y) = \cos x \cos y + \sin x \sin y$. Adding: the $\sin x \sin y$ terms cancel, leaving $2\cos x \cos y$.
- 8 Verify the identity $\sin(x + y) - \sin(x - y) = 2\cos x \sin y$.
 $\sin(x + y) = \sin x \cos y + \cos x \sin y$ and $\sin(x - y) = \sin x \cos y - \cos x \sin y$. Subtracting: the $\sin x \cos y$ terms cancel, leaving $2\cos x \sin y$.
- 9 Show that $\tan \left(x + \frac{\pi}{4} \right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.
With $B = \frac{\pi}{4}$, $\tan B = 1$: $\tan \left(x + \frac{\pi}{4} \right) = \frac{\tan x + 1}{1 - \tan x \cdot 1} = \frac{1 + \tan x}{1 - \tan x}$.

Applications

- 10** Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

$\sin\left(x + \frac{\pi}{6}\right) = \sin x \cos \frac{\pi}{6} + \cos x \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x$. The coefficient of $\cos x$ is $\frac{1}{2}$.

Additional practice

- 11** Find the exact value of $\cos \frac{\pi}{12}$.

$$\cos \frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

- 12** Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{3/4 + 5/12}{1 - (3/4)(5/12)} = \frac{14/12}{1 - 15/48} = \frac{7/6}{33/48} = \frac{7}{6} \cdot \frac{48}{33} = \frac{56}{33}.$$

- 13** Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

This matches the pattern $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = \frac{\pi}{2} - x$, $B = x$: $= \cos\left(\frac{\pi}{2} - 2x\right)$.